



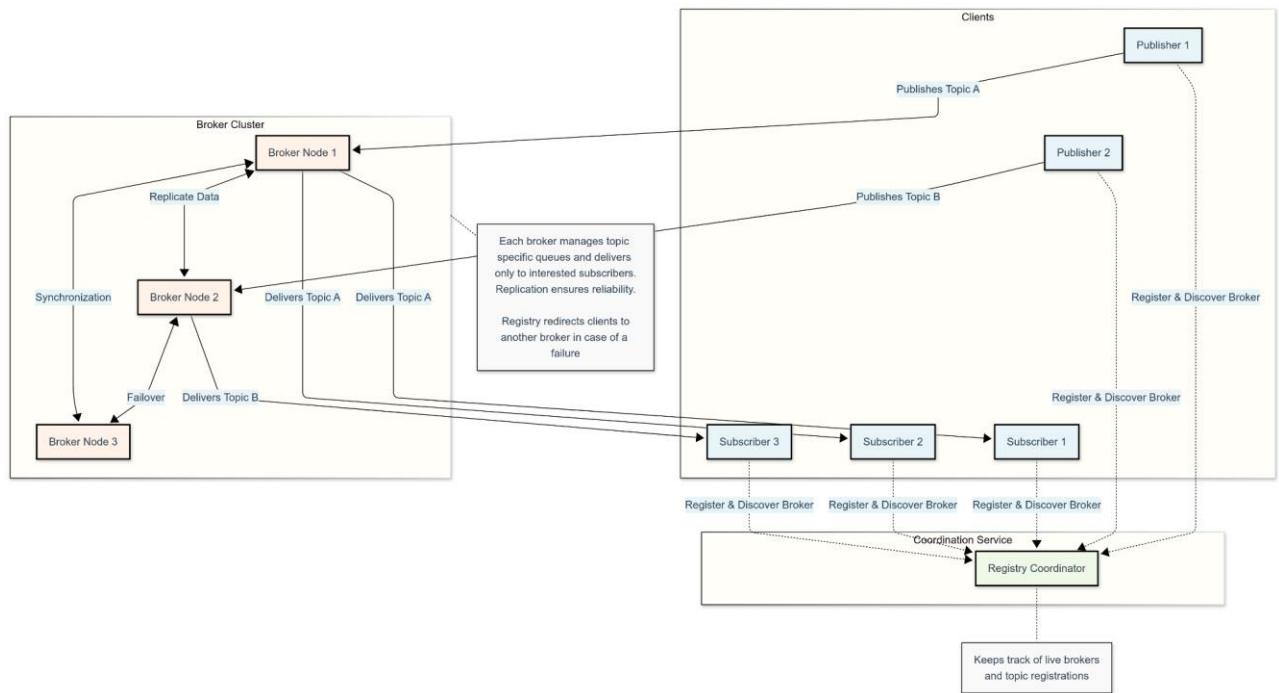
Middleware Architectures [IS3108 / SCS3203]

Group Assignment

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Proposed distributed architecture for the Pub/Sub implementation



Description of Architecture Enhancements

1. Broker Cluster (B1, B2, B3):

- Multiple broker servers run concurrently to eliminate a single point of failure.
- Each broker handles a subset or full set of topics, with synchronization across brokers.

2. Coordination Service (Registry/Coordinator):

- Maintains metadata about:
 - Active brokers
 - Topics managed by each broker
 - Publisher/subscriber registrations
 - Supports failover and dynamic re-routing of clients to available brokers.

3. Clients (Publishers/Subscribers):

- a. First, contact the coordination service to find a broker.
- b. Connect to an assigned broker for Pub/Sub actions.

4. Failover Handling:

- a. If a broker crashes, the coordinator informs clients to reconnect to another healthy broker.
- b. Brokers sync data periodically to ensure message continuity

5. Fault Tolerance:

- a. Brokers use the shared registry to recover subscriptions and topics after a crash.

6. Message Persistence:

- a. Shared storage ensures that messages and subscriptions are not lost even if a broker node fails.

7. Load Balancing:

- a. Incoming connections are distributed across brokers.
- b. Ensures even workload distribution and better performance.

8. Scalability:

- a. Can horizontally scale by adding more brokers.
- b. Supports more publishers/subscribers without performance degradation.

9. High Availability:

- a. Multiple broker nodes eliminate single points of failure.
- b. If one broker crashes, others continue handling traffic.

Drive link for the task videos: -

https://drive.google.com/drive/folders/1LDj4kn1PQpPj_sJlAWoJ3z3_f08BiS3v?usp=sharing